

# **PROJECT REPORT**

## **Composition-Dependent Elastic Properties of the Fe–C System**

Analysis of Young's Modulus,  $C_{11}$ (Elastic stiffness constant) and Shear Modulus  
as a Function of Carbon Content

Prepared based on the results of thermocalc plots

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## 1. Introduction

Iron–carbon alloys form the basis of a large class of engineering materials, particularly steels and cast irons. Carbon strongly affects phase stability, microstructure and mechanical behaviour. Although hardness, strength and ductility are commonly discussed for Fe–C alloys, elastic properties are also important because they govern the initial deformation response of a material under load.

In a computational materials approach, elastic properties can be evaluated as a function of composition and phase stability. The supplied file contains six plots generated for the Fe–C composition range from 0 to 100 mass % C. The figures show Young’s modulus, C11 and shear modulus for the phases and phase combinations represented in the calculation. Such plots allow the effect of carbon composition and phase constitution on stiffness-related properties to be examined without performing a separate experiment for every composition.

## 2. Problem Statement

To examine how carbon concentration influences the calculated elastic properties of the Fe–C system and to interpret the resulting trends in terms of the phases present.

## 3. Objectives

- To study the variation of Young’s modulus with mass percent carbon.
- To examine the variation of the elastic constant C11 with carbon concentration.
- To analyse the variation of shear modulus across the Fe–C composition range.
- To identify the phase fields represented by BCC\_A2, FCC\_A1, GRAPHITE\_A9 and liquid.
- To compare the qualitative stiffness response of different phase combinations.
- To understand the usefulness and limitations of computational elastic-property predictions.

## 4. Theoretical Background

### 4.1 Young’s Modulus

Young’s modulus ( $E$ ) is a measure of the resistance of a material to elastic deformation in tension or compression. For a linear elastic material,  $E$  is defined as the ratio of normal stress to normal strain. A higher Young’s modulus indicates a stiffer response. The unit used in the supplied plots is GPa.

### 4.2 Elastic Constant C11

C11 is one of the independent elastic stiffness constants used to describe the response of a crystal to strain. For cubic crystal systems such as BCC and FCC, C11 is one of the principal single-crystal elastic constants, along with C12 and C44. Its value is useful for understanding directional stiffness at the crystal level.

### 4.3 Shear Modulus

The shear modulus ( $G$ ) represents the resistance of a material to shear deformation. A larger shear modulus means that greater shear stress is required to produce a given elastic shear strain. The supplied plots report shear modulus in GPa.

## 4.4 Phases Represented in the Results

| Phase / notation | Meaning                                 | Role in the calculation                                           |
|------------------|-----------------------------------------|-------------------------------------------------------------------|
| BCC_A2           | Body-centred cubic solid solution phase | Iron-rich solid phase represented in the plots                    |
| FCC_A1           | Face-centred cubic solid solution phase | Austenitic-type solid phase represented in the plots              |
| GRAPHITE_A9      | Graphite phase                          | Carbon-rich solid phase                                           |
| LIQUID           | Liquid phase                            | High-temperature / molten state represented in phase combinations |
| Two-phase fields | Combinations such as BCC_A2 + FCC_A1    | Properties associated with the calculated phase assemblage        |

## 5. Methodology

The analysis was performed using the computational results contained in the supplied Pages document. The calculation covers a composition axis from 0 to 100 mass percent carbon. The output quantities were plotted against carbon concentration for the phases and phase combinations included in the calculation.

1. Define the Fe–C composition range on a mass-percent-carbon basis.
2. Evaluate the stable/represented phases and their combinations over the composition range.
3. Calculate the elastic-property quantities associated with the phases.
4. Generate composition-dependent plots for Young’s modulus, C11 and shear modulus.
5. Compare the trends and identify regions where the elastic response changes significantly.

The present report interprets the supplied plots directly. Numerical values are discussed qualitatively because the source document contains graphs but does not provide the underlying tabulated numerical dataset.

## 6. Results and Discussion

### 6.1 Young’s Modulus

The Young’s modulus plots show a strong dependence on carbon concentration. At the carbon-rich end of the composition axis, the curves converge toward very small values. In the iron-rich region, substantially higher values are obtained, with several phase-dependent curves extending to approximately the 150–225 GPa range in the supplied plots. The separation between curves indicates that the predicted stiffness depends not only on composition but also on the phase constitution.

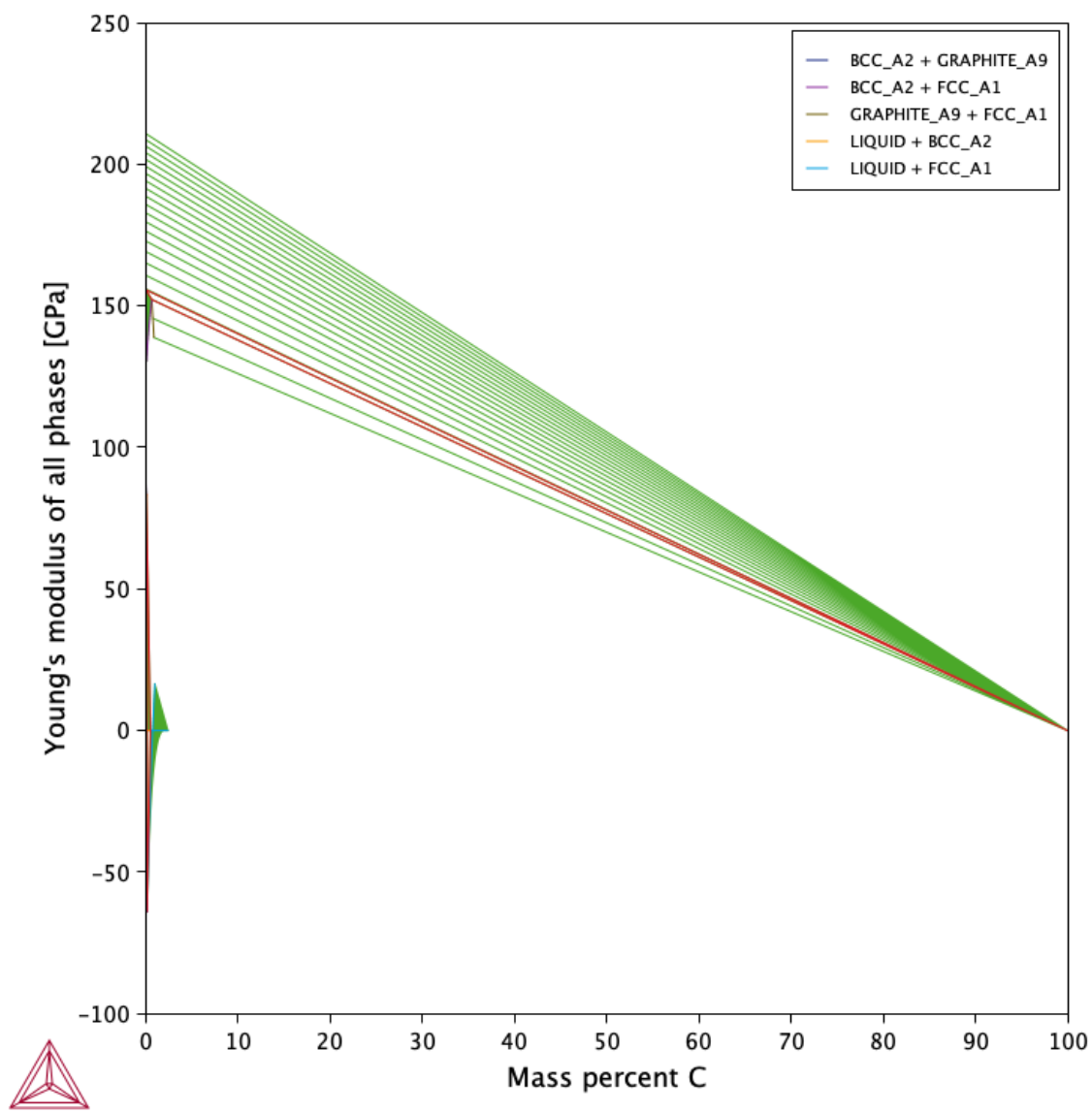


Figure 1. Young's modulus of the represented phases/phase combinations versus mass percent C (first calculation set).

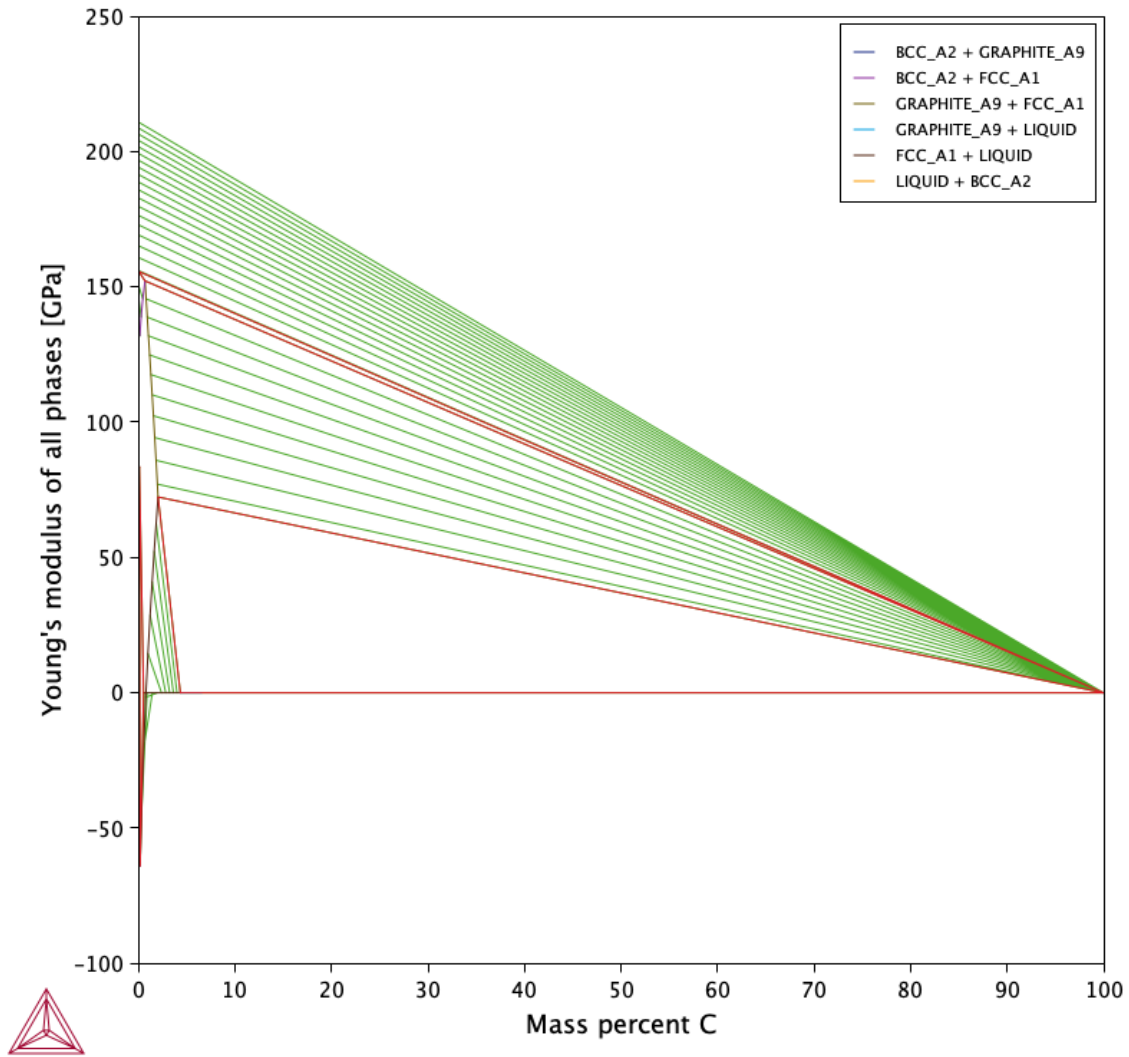


Figure 2. Young's modulus versus mass percent C (second calculation set).

## 6.2 Elastic Constant C11

The C11 results follow the same broad composition-dependent tendency: the calculated stiffness constant is much higher on the iron-rich side and decreases as the composition approaches pure carbon. The iron-rich region shows a family of curves, reflecting different phase/property combinations. The carbon-rich end approaches zero in the displayed results.

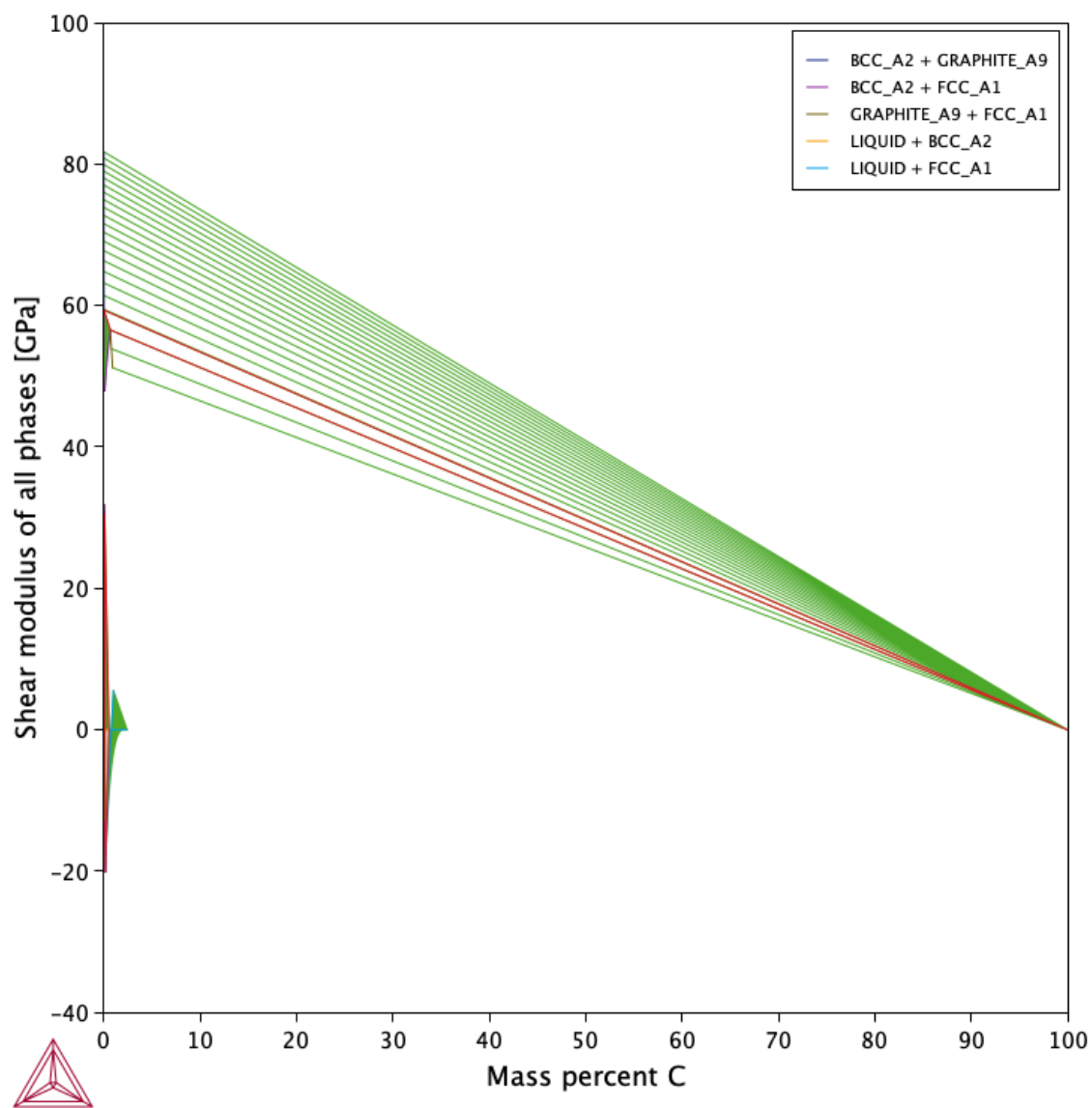


Figure 3. C11 versus mass percent C (first calculation set).

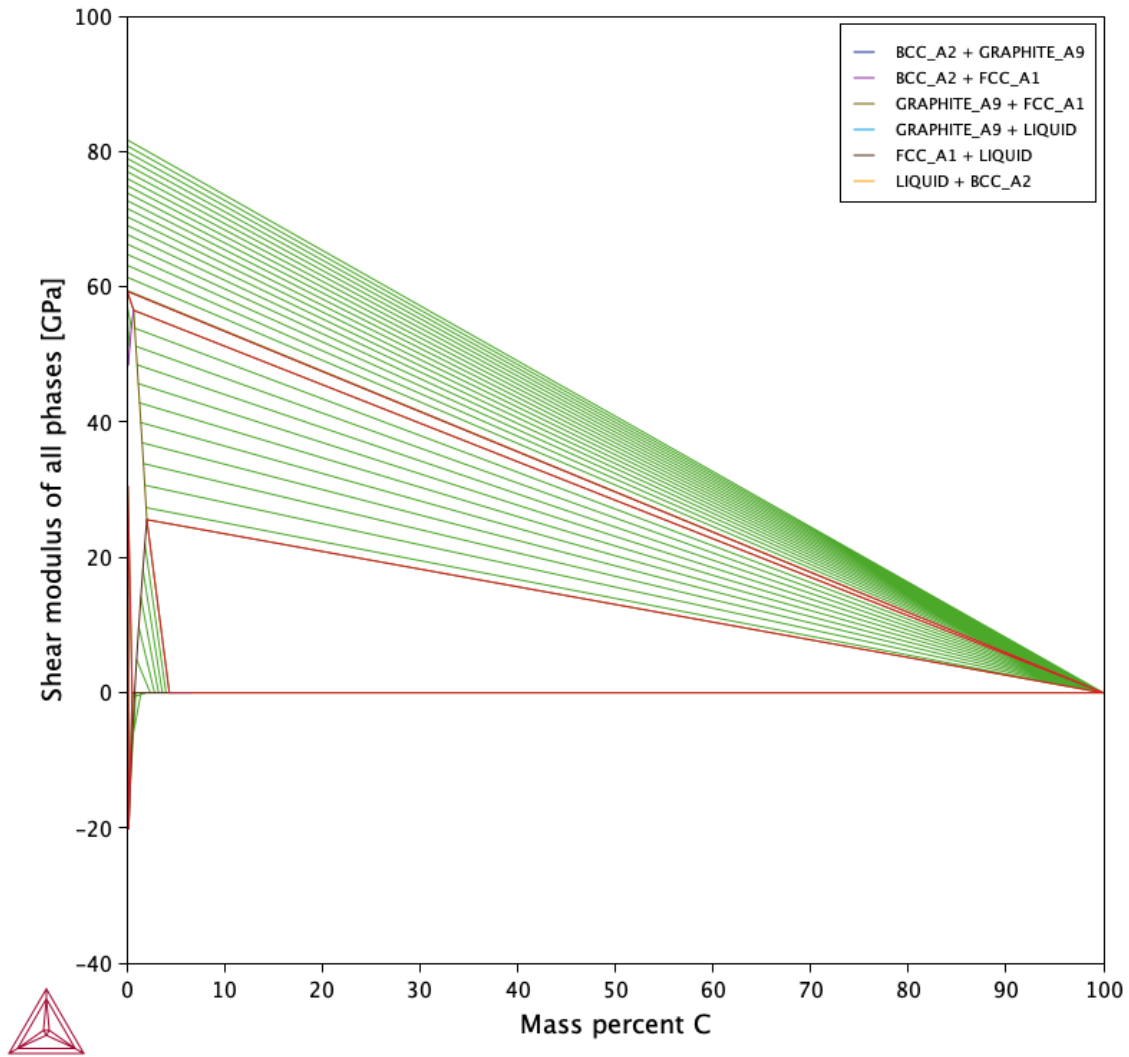
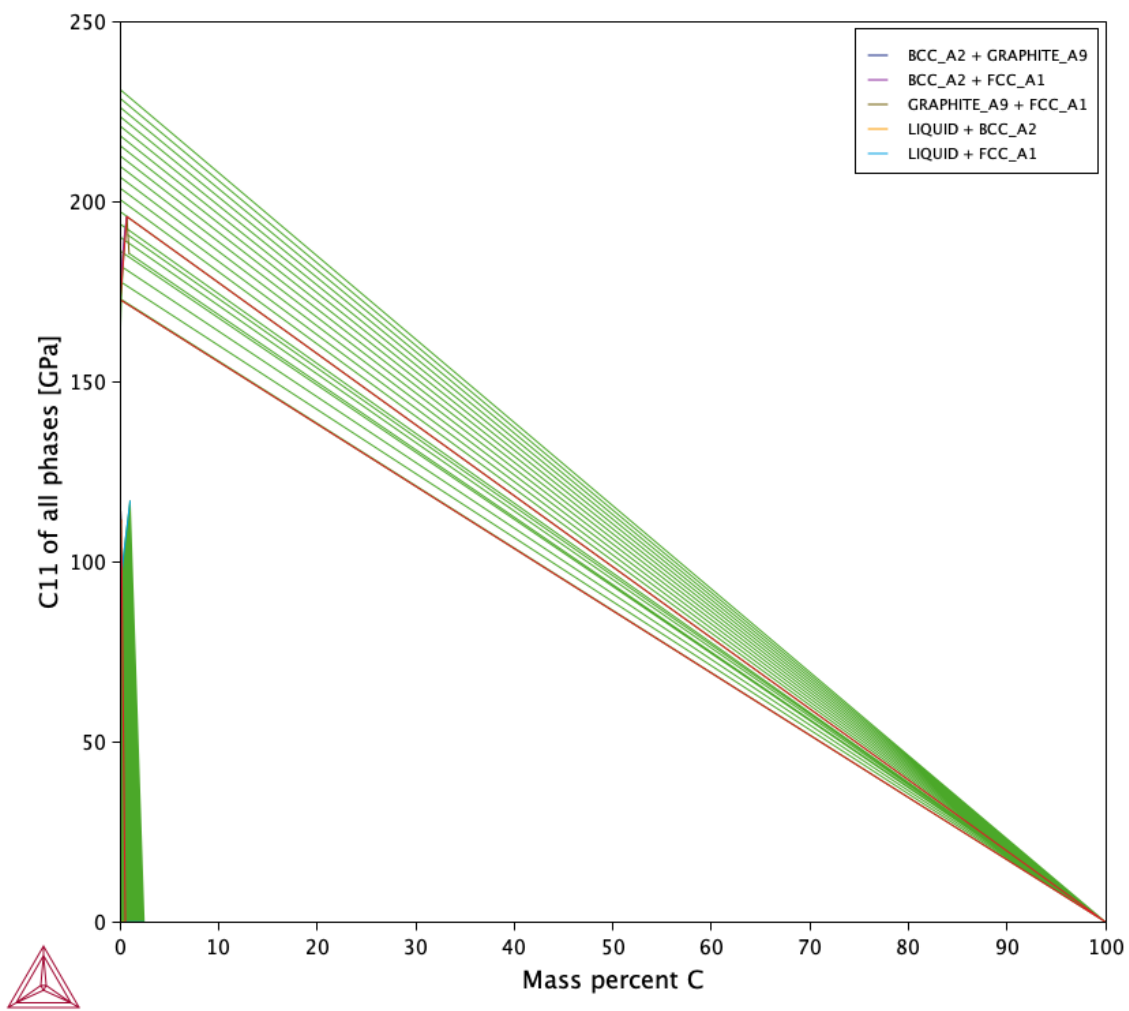


Figure 4. C11 versus mass percent C (second calculation set).

### 6.3 Shear Modulus

The shear-modulus plots also exhibit a decreasing trend with increasing carbon content. In the iron-rich region, the calculated shear modulus reaches considerably higher values than in the carbon-rich region. The curves near the low-carbon end show more pronounced phase-dependent behaviour, while most curves converge toward zero as the composition approaches 100 mass % C.



**Figure 5. Shear modulus versus mass percent C (first calculation set).**

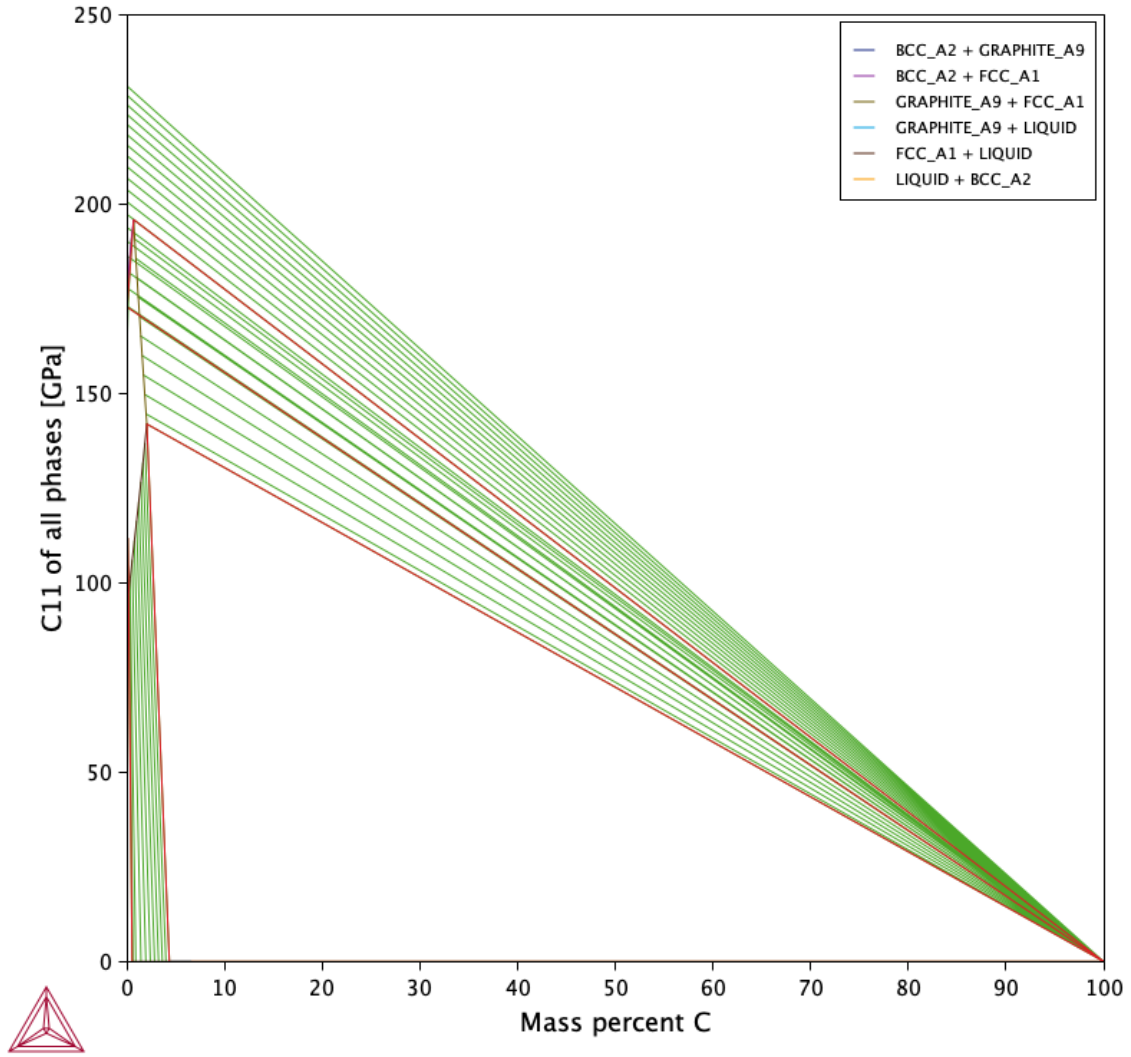


Figure 6. Shear modulus versus mass percent C (second calculation set).

#### 6.4 Overall Comparison

Across the supplied figures, Young's modulus, C11 and shear modulus show a common qualitative trend: the plotted stiffness quantities are highest in the Fe-rich portion and decline strongly toward the C-rich end. This is expected to be closely related to the change in phase constitution and the intrinsic elastic response of the phases involved. The presence of BCC\_A2, FCC\_A1 and graphite-related phase fields also demonstrates that a simple composition-only interpretation is insufficient; the phase state must be considered when evaluating material properties.

## 7. Observations

| Property           | Observed trend                                           | Interpretation                                                          |
|--------------------|----------------------------------------------------------|-------------------------------------------------------------------------|
| Young's modulus    | Generally decreases toward high C content                | Elastic stiffness is strongly composition/phase dependent               |
| C11                | High on Fe-rich side and approaches low values at high C | Crystal-level stiffness changes with phase and composition              |
| Shear modulus      | Generally decreases with increasing C                    | Resistance to shear deformation becomes lower in the carbon-rich region |
| Phase constitution | BCC_A2, FCC_A1, graphite and liquid combinations appear  | Elastic response is linked to phase stability and phase assemblage      |

## 8. Engineering Significance

Elastic properties are important in component design, finite-element modelling, vibration analysis, thermal stress calculations and the interpretation of mechanical tests. Composition-dependent computational results can therefore be useful for selecting candidate compositions and for generating material-property inputs for numerical simulations.

For steels, carbon is normally present at relatively low concentrations compared with the full 0–100 mass % C range shown in these plots. Consequently, the most practically relevant part of the curves for conventional steels is the iron-rich, low-carbon region. The high-carbon portion is still useful for understanding the limiting behaviour of the Fe–C system and for studying carbon-rich phases such as graphite.

## 9. Limitations

- The supplied document contains plots but not the complete numerical data table, so exact values cannot be reproduced from the file alone.
- The plots do not explicitly state all calculation conditions in the visible figure area; therefore temperature, pressure and database settings should be recorded separately when reporting final numerical results.
- Calculated elastic properties depend on the thermodynamic/property database and modelling assumptions used.
- Phase-assemblage properties should not automatically be interpreted as experimental bulk properties without checking the homogenization/model used.
- Experimental measurements are required to validate the predictions for a particular alloy, processing condition and microstructure.

## 10. Conclusion

The project successfully examines the composition dependence of three important elastic-property quantities—Young's modulus, C11 and shear modulus—in the Fe–C system. The supplied computational plots show that all three properties vary strongly with carbon concentration and phase constitution. The highest stiffness

values occur predominantly in the iron-rich region, while the plotted values decrease substantially toward the carbon-rich end. The results also demonstrate the importance of distinguishing between BCC\_A2, FCC\_A1, graphite and liquid-related phase fields when interpreting material properties.

Overall, the work illustrates how computational materials engineering can be used to investigate composition–property relationships efficiently. The next step would be to export the underlying numerical data, specify the exact thermodynamic/property database and calculation conditions, and compare the predictions with experimental elastic-property data for selected Fe–C alloys.

## 11. Future Work

- Export numerical property data at selected carbon concentrations for quantitative comparison.
- Repeat the calculation at multiple temperatures to study temperature-dependent elastic behaviour.
- Compare calculated properties with experimental data for ferrite, austenite, steels and cast irons.
- Investigate the effect of other alloying elements such as Mn, Si, Cr and Ni.
- Use the predicted elastic constants as inputs for computational mechanics or finite-element simulations.

## 12. References / Recommended Sources

6. Callister, W. D. and Rethwisch, D. G., Materials Science and Engineering: An Introduction.
7. Dieter, G. E., Mechanical Metallurgy.
8. ASM Handbook, Volume 1: Properties and Selection: Irons, Steels, and High-Performance Alloys.
9. Thermo-Calc documentation and database manuals corresponding to the property calculations used to generate the supplied figures.

**End of Report**